

Metric Definitions and Fine-Grid Sensitivity Analysis

CUIMC Appointment Simulation

May 7, 2026

1 Purpose

This document defines the final reporting metrics and analyzes how the simulation responds to class-specific parameter changes. The figures in this version use finer grids than the first analysis:

- Balking and no-show step probabilities: 21 values from 0.00 to 1.00.
- Cancellation probabilities: 21 values from 0.00 to 0.30.
- Thresholds: every feasible integer threshold in the baseline booking horizon.
- Common threshold-jump sweeps: every feasible threshold crossed with 21 jump levels from 0.00 to 1.00.
- Arrival mix: 11 total-arrival multipliers and 17 class-mix values.
- FCFS capacity stress: 15 total-arrival multipliers from 0.30 to 1.70, preserving the baseline 50/50 class mix.
- Regression screening: 240 randomized FCFS parameter settings, averaged over 2 random seeds each.

The focus is not only aggregate performance. We also ask whether one class is consistently benefited relative to the other, and whether the effect is caused by an absolute parameter level or by the difference between class 1 and class 2.

2 Metric and Variable Definitions

Symbol	Code name	Definition
i	<code>class_id</code>	Patient class, currently $i \in \{1, 2\}$.
d	<code>measured day</code>	Day inside the measurement window.
D	<code>measure_days</code>	Number of measured days.
S	<code>slots_per_day</code>	Daily clinic appointment capacity.
τ	<code>booking delay</code>	Offered delay in days between arrival and appointment date.
A_i	<code>arrivals</code>	Measured arrivals from class i .
B_i	<code>booked</code>	Class- i arrivals that accepted an offered slot.
K_i	<code>balked</code>	Class- i arrivals that received an offer but rejected it.
N_i	<code>no_offer</code>	Class- i arrivals that did not receive any slot offer.
C_i	<code>canceled</code>	Class- i booked patients who canceled before service.
H_i	<code>no_show</code>	Class- i booked patients who did not attend.
V_i	<code>served</code>	Class- i patients who completed service.
O_i	<code>offered</code>	Class- i patients who received an offer: $O_i = B_i + K_i$.

Table 1: Notation used in metric definitions.

Patients with `no_offer` are not included in offered-delay metrics because they did not receive a slot offer.

Metric	Code name	Formula	Interpretation
Accepted delay, class i	<code>mean_accepted_booking_delay_i</code>	$\frac{\sum_{\text{accepted class } i} \tau}{B_i}$	Average delay among class- i patients who accepted/booked a slot.
Accepted delay, aggregate	<code>mean_accepted_booking_delay</code>	$\frac{\sum_i \sum_{\text{accepted class } i} \tau}{\sum_i B_i}$	Average delay among all accepted/booked patients. This is the old <code>mean_booking_delay</code> alias.
Offered delay, class i	<code>mean_offered_booking_delay_i</code>	$\frac{\sum_{\text{offered class } i} \tau}{O_i}$	Average delay shown to class- i patients who received an offer, including patients who balked. Excludes <code>no_offer</code> .
Offered delay, aggregate	<code>mean_offered_booking_delay</code>	$\frac{\sum_i \sum_{\text{offered class } i} \tau}{\sum_i O_i}$	Average offered delay across all patients who received an offer.
Percent serviced, class i	<code>percent_serviced_i</code>	$\frac{V_i}{A_i}$	Main class-level access metric.
Percent serviced, aggregate	<code>overall_percent_serviced</code>	$\frac{\sum_i V_i}{\sum_i A_i}$	Main aggregate access metric: share of arrivals that became completed visits.
Average utilization	<code>average_utilization</code>	$\frac{1}{D} \sum_{d=1}^D \frac{\text{served_slots}_d}{S}$	Average daily share of capacity used by completed visits. This is aggregate only; no-shows do not count as utilized slots.

Table 2: Primary reporting metrics.

Metric or diagnostic	Formula / definition	Interpretation
Δ_{access}	<code>percent_serviced₁</code> – <code>percent_serviced₂</code>	Positive means class 1 is served at a higher rate; negative means class 2 is served at a higher rate.
Δ_{delay}	<code>mean_offered_booking_delay₂</code> – <code>mean_offered_booking_delay₁</code>	Positive means class 1 receives shorter offered delays; negative means class 2 receives shorter offered delays.
<code>booked_slots</code>	Slots that had a booked appointment at service time, including no-shows	Diagnostic count only, not final utilization.
<code>served_slots</code>	Booked slots that became completed visits	Numerator for utilization.
<code>no_show_slots</code>	Booked slots lost to no-shows	Diagnostic count for lost capacity.

Table 3: Class-benefit metrics and slot diagnostics. `empty_slots` and `empty_slot_rate` are not used for final reporting.

3 Scenario Comparison

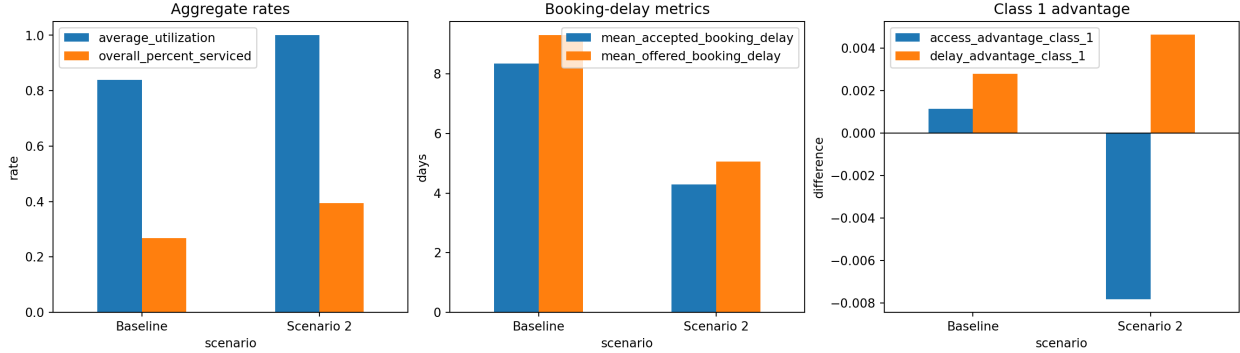


Figure 1: Scenario comparison across aggregate rates, booking delays, and class-1 advantage metrics.

Scenario	Avg. util.	Overall svc.	Accepted delay	Offered delay	Δ_{access}	Δ_{delay}
Baseline	0.839	0.269	8.35	9.30	0.001	0.003
Scenario 2	1.000	0.395	4.29	5.06	-0.008	0.005

Table 4: Scenario-level metrics. Scenario values are rates or averages, not total counts.

The baseline has almost no class advantage because the two classes are symmetric. Scenario 2 improves aggregate performance, but that improvement is not caused by one class being structurally favored. It comes from absolute system changes: lower total demand, shorter balking threshold, stronger balking after threshold, and a no-show threshold that avoids no-shows within the available booking horizon.

4 How to Read the Heatmaps

Each sensitivity heatmap has four panels:

- Average utilization: capacity used by completed visits.
- Overall percent serviced: aggregate access.
- Class 1 access advantage: Δ_{access} .
- Class 1 delay advantage: Δ_{delay} .

For the two advantage panels, red/blue color indicates direction around zero. Positive values favor class 1. Negative values favor class 2. The diagonal of a class-vs-class heatmap is especially important: on the diagonal, the two classes have the same parameter value. Any strong class advantage away from the diagonal is therefore caused by a difference between class parameters, not by the absolute parameter level alone.

The threshold-jump heatmaps are different: both classes receive the same threshold and the same jump level. Those plots isolate absolute system effects. Any class advantage in those panels should stay near zero unless simulation noise or volume imbalance is present.

Some balking heatmaps include a dashed horizontal row. That row is the one-dimensional slice shown immediately below the heatmap: class 1 varies along the x-axis while class 2 is fixed at its baseline value.

5 Balking

5.1 Step Sensitivity

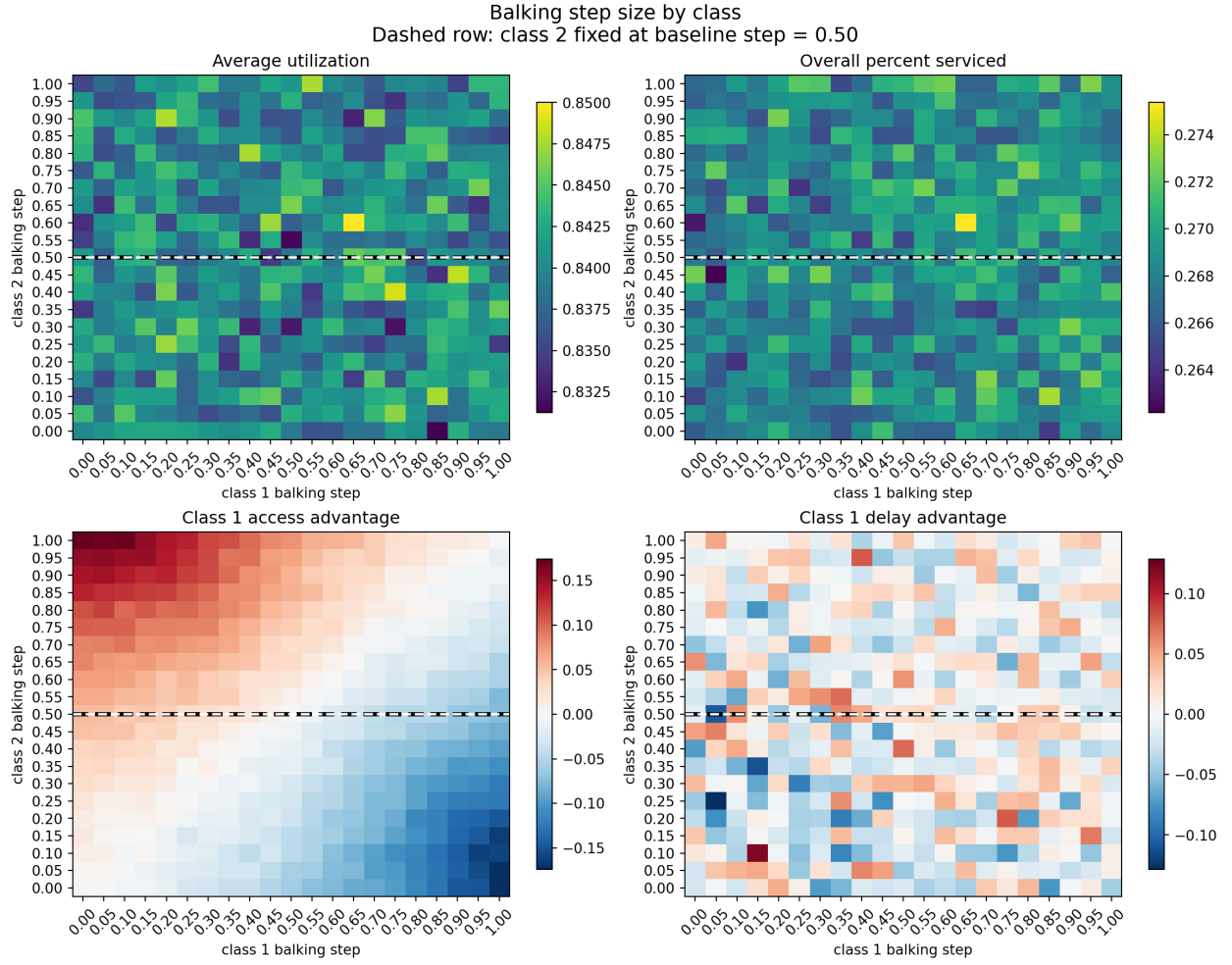


Figure 2: Fine-grid balking step sensitivity by class. The dashed row fixes class 2 at the baseline balking step, 0.50.

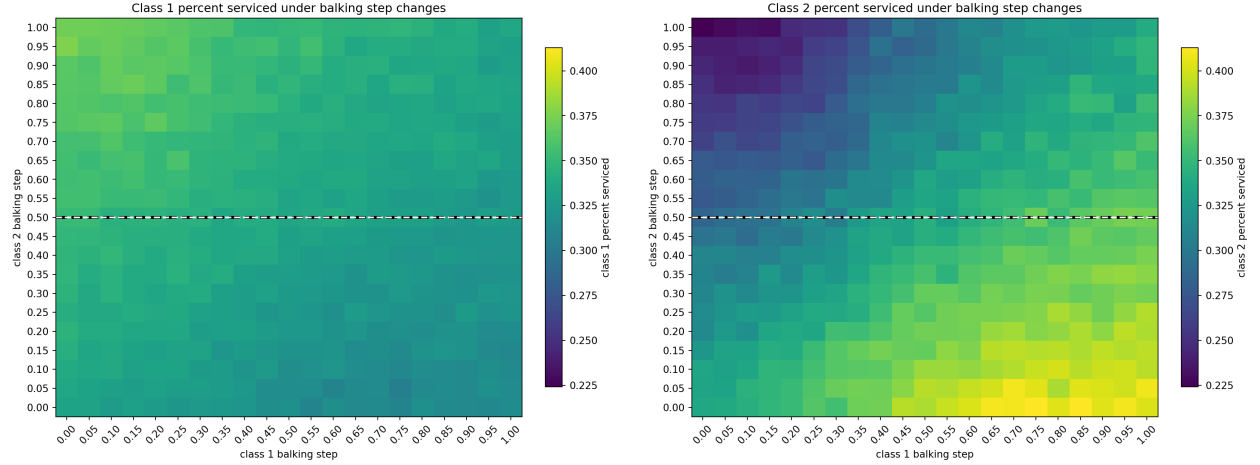


Figure 3: Class-specific service heatmaps for the balking step sweep. Left: class 1 percent serviced. Right: class 2 percent serviced. Both panels use the same class-1-step and class-2-step axes and the same color scale.

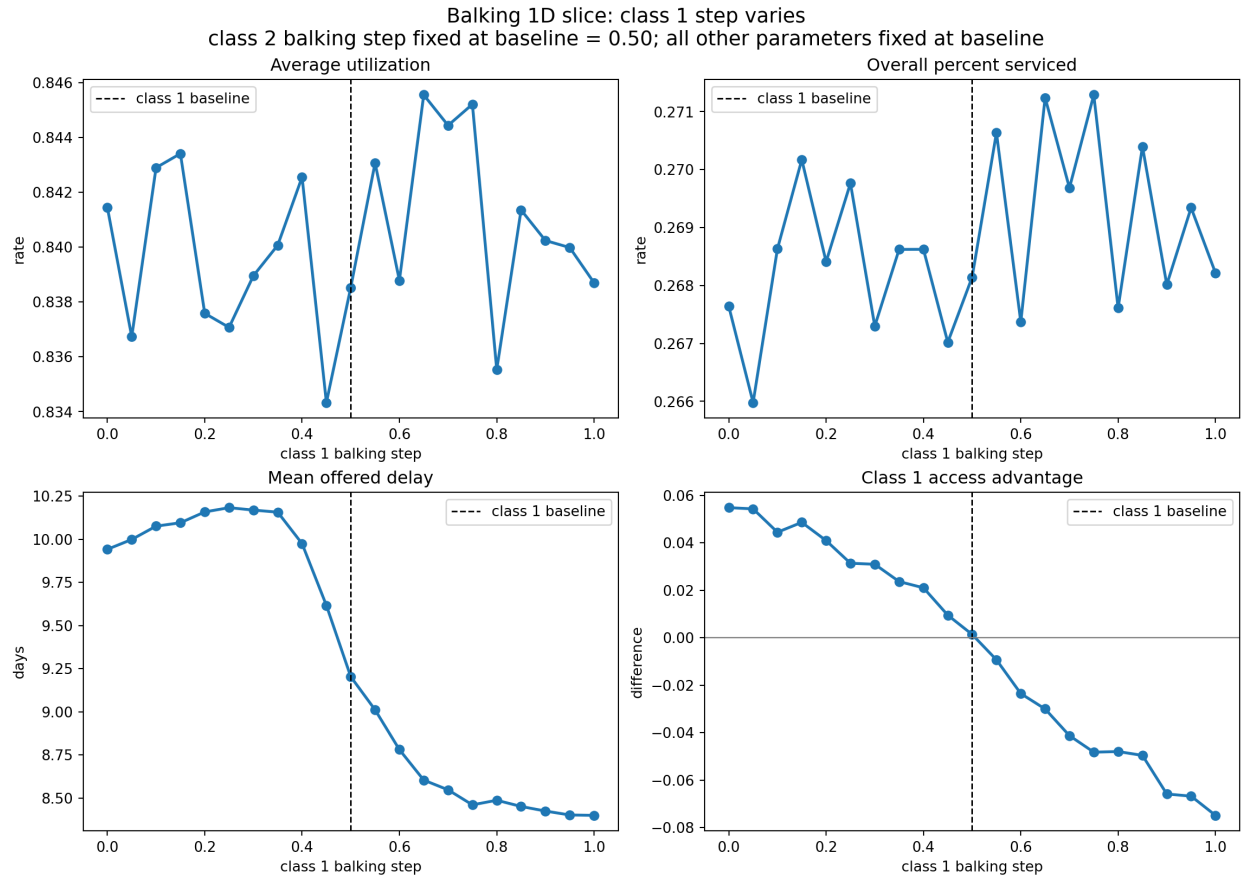


Figure 4: One-dimensional slice of the balking step heatmap. Class 1 balking step varies; class 2 balking step is fixed at the baseline value 0.50; all other parameters stay at baseline.

The balking step is the probability jump after the balking threshold. With baseline $\text{low} = 0$, the step is the high-delay balking probability.

Aggregate impact. Aggregate utilization and overall percent serviced change modestly across the balking-step grid. Balking mostly changes who accepts long-delay offers, not the physical capacity of the clinic.

Class-benefit dynamic. The class with the lower balking step is benefited in access. If class 1 has low balking and class 2 has high balking, class 1 has a much higher percent serviced. If the pattern is reversed, class 2 is benefited. In the fine grid, the largest access advantage is about 0.174 in absolute value. In the one-dimensional slice, fixing class 2 at 0.50 makes this direction explicit: class 1 is benefited when its step is below 0.50, and class 2 is benefited when class 1's step is above 0.50.

Interpretation. Balking asymmetry is a class-benefit parameter. Absolute balking level affects delay and access tradeoffs, but the difference between class-specific balking steps determines which class is served more often. The class-specific service heatmaps show this directly: each class's serviced share is highest when that class is less likely to balk and the other class is more likely to balk.

5.2 Threshold Sensitivity

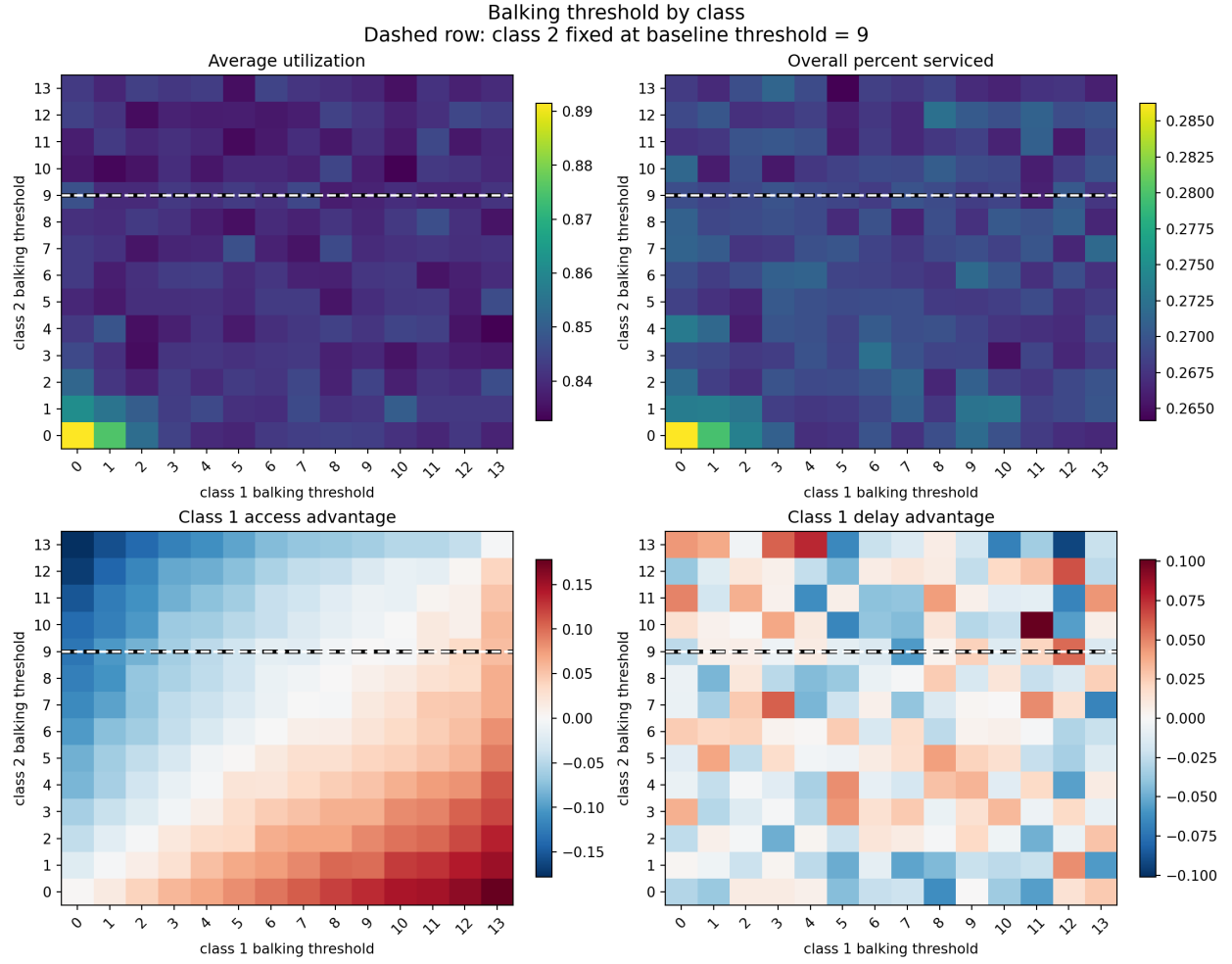


Figure 5: Balking threshold sensitivity by class. The dashed row fixes class 2 at the baseline balking threshold, 9 days.

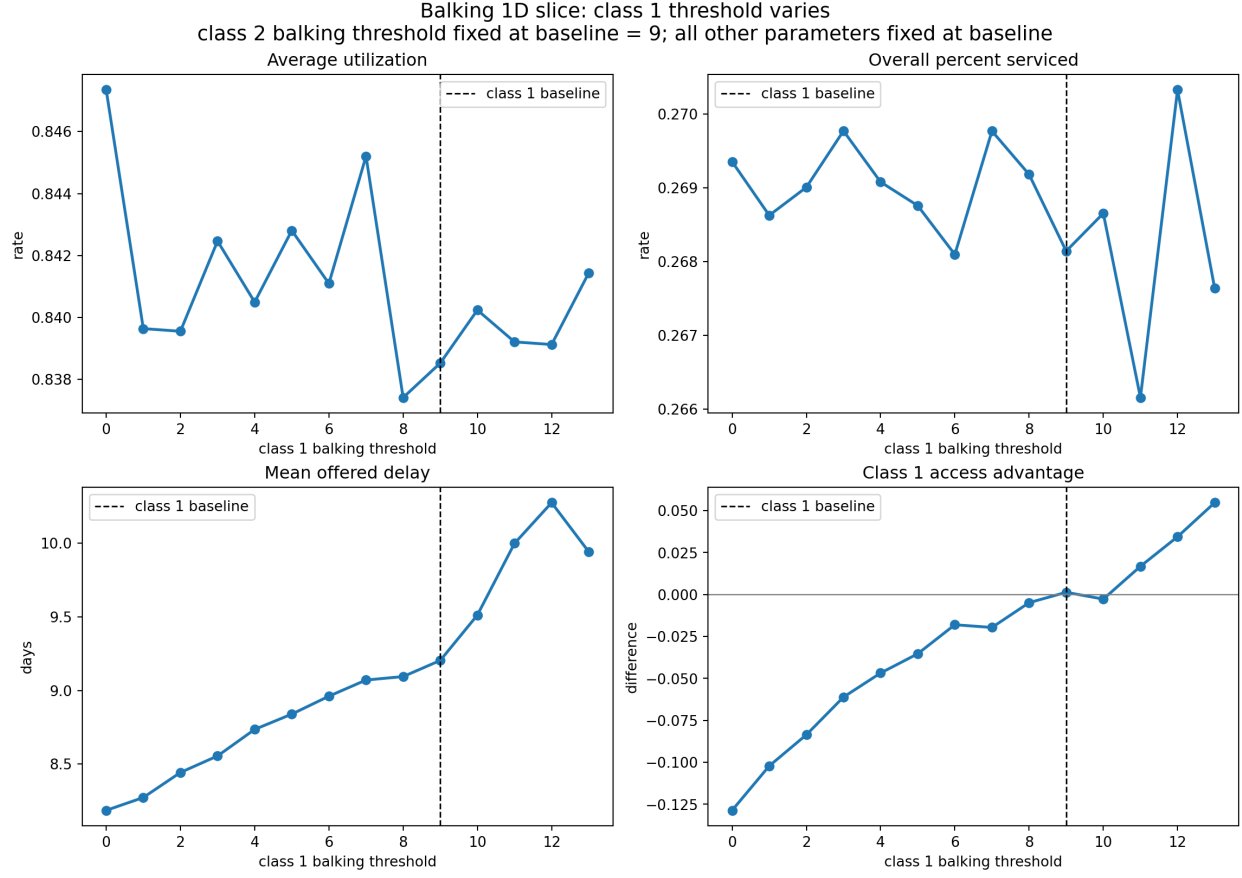


Figure 6: One-dimensional slice of the balking threshold heatmap. Class 1 balking threshold varies; class 2 balking threshold is fixed at the baseline value 9 days; all other parameters stay at baseline.

The balking threshold is the delay at which the high balking probability begins. A higher threshold means patients tolerate longer offered delays before entering the high-balking region.

Aggregate impact. The absolute threshold level changes delay. High thresholds allow more long-delay offers to be accepted, increasing accepted/offered delay.

Class-benefit dynamic. The class with the higher balking threshold is benefited in access because its patients tolerate longer waits. When class 1 has a high threshold and class 2 has a low threshold, class 1 is benefited; when reversed, class 2 is benefited. The largest access advantage is about 0.178 in absolute value. In the one-dimensional slice, class 2 is fixed at threshold 9: class 1 is disadvantaged below that value and increasingly benefited above it.

Interpretation. Balking threshold differences behave similarly to balking step differences: relative tolerance across classes determines which class is advantaged.

5.3 Threshold and Jump Level

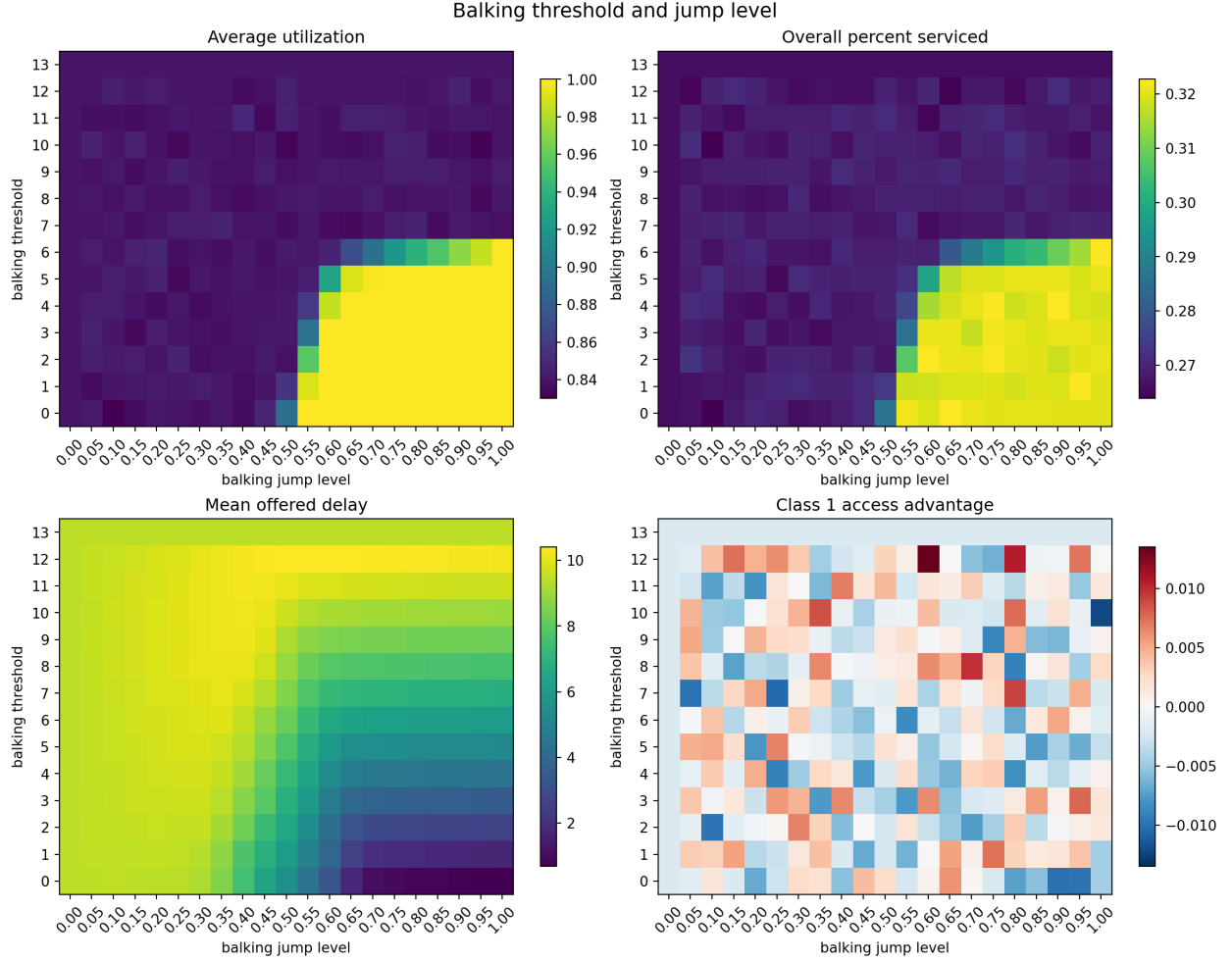


Figure 7: Balking sensitivity when both classes share the same threshold and jump level.

This sweep varies the absolute balking threshold and the absolute balking jump level together. Both patient classes use the same values at each grid point.

Aggregate impact. The main effect is a delay/access tradeoff. Higher thresholds and lower jump levels make patients more willing to accept long-delay offers, so offered delay increases. Lower thresholds and higher jump levels reject more long-delay offers, which can reduce delay among offered/accepted patients but also reduce access.

Class-benefit dynamic. Because the two classes share the same balking rule in this sweep, there is no structural class benefit. The largest observed access advantage is only about 0.014 in absolute value, much smaller than the class-specific balking sweeps.

Interpretation. The absolute balking threshold and jump level matter for system-level delay and access. A class is benefited only when its balking rule differs from the other class.

6 No-Show

6.1 Step Sensitivity

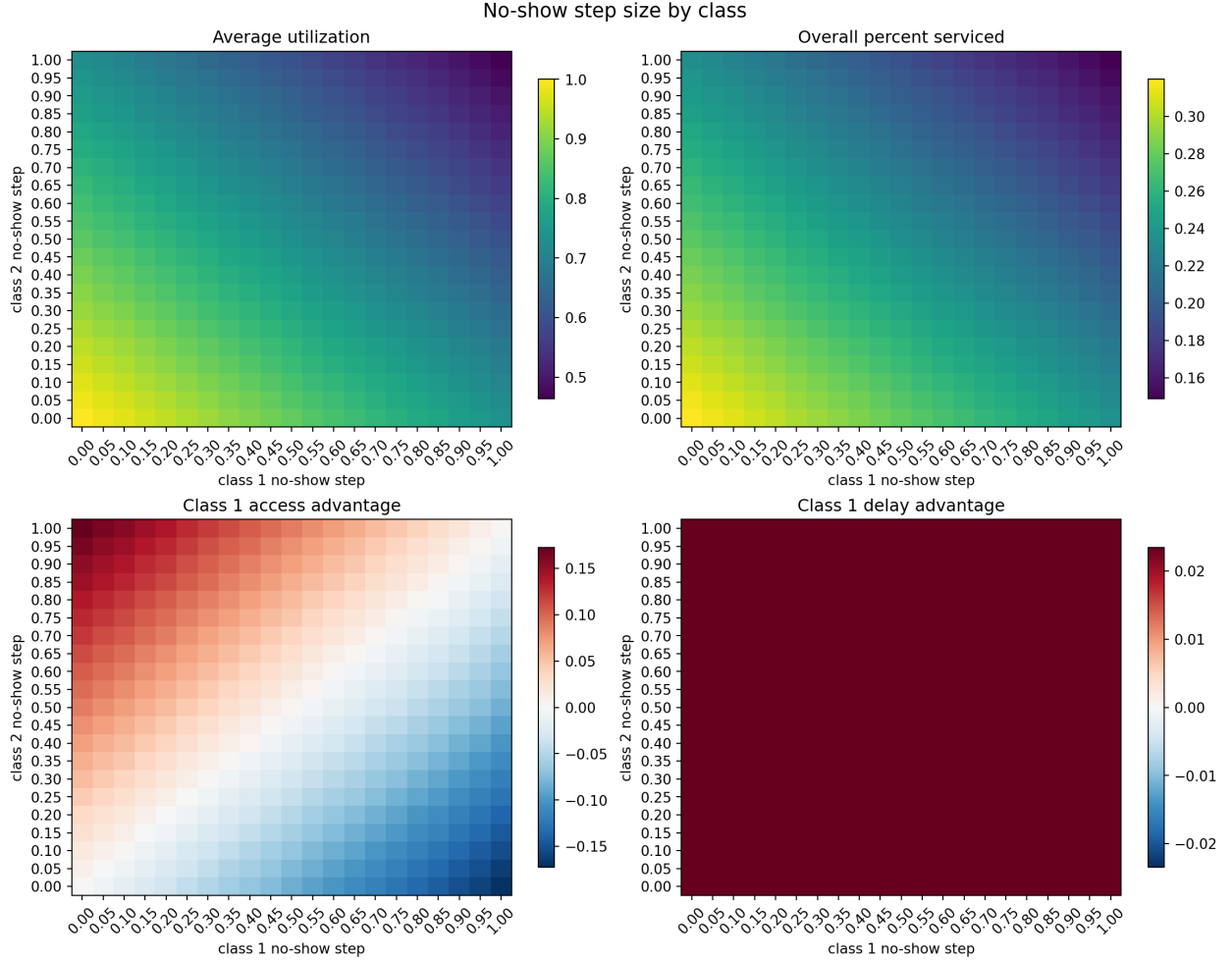


Figure 8: Fine-grid no-show step sensitivity by class.

The no-show step is the probability jump after the no-show threshold.

Aggregate impact. This is one of the strongest aggregate utilization drivers. Higher no-show probability directly converts booked capacity into lost visits.

Class-benefit dynamic. The class with the lower no-show step is benefited in percent serviced. The access advantage reaches about 0.172 in absolute value. However, the delay advantage barely changes because no-shows occur after booking. No-show behavior changes whether a booked patient becomes a completed visit, but not the delay that was offered when the appointment was booked.

Interpretation. No-show asymmetry can strongly benefit one class in service completion, but it is not a booking-delay lever.

6.2 Threshold Sensitivity

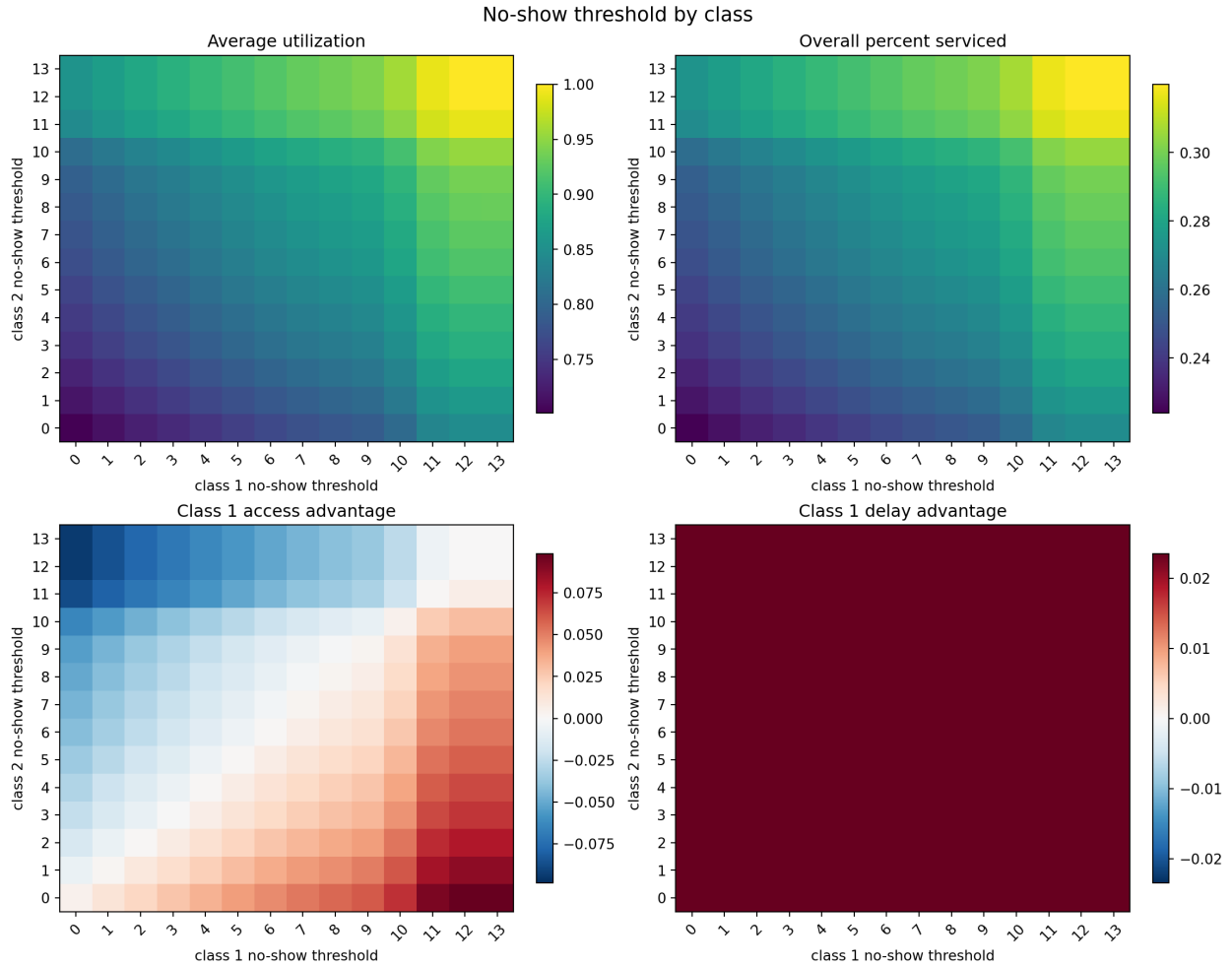


Figure 9: No-show threshold sensitivity by class.

A higher no-show threshold means no-shows only increase after longer booking delays.

Aggregate impact. Raising thresholds for both classes improves utilization because fewer booked patients no-show.

Class-benefit dynamic. The class with the higher no-show threshold is benefited in access. The maximum access advantage is smaller than for the full step-size sweep, about 0.098, because the high no-show probability itself remains fixed.

Interpretation. No-show threshold differences matter, but less than no-show step differences. Thresholds determine when the risk begins; step size determines how severe the risk becomes.

6.3 Threshold and Jump Level

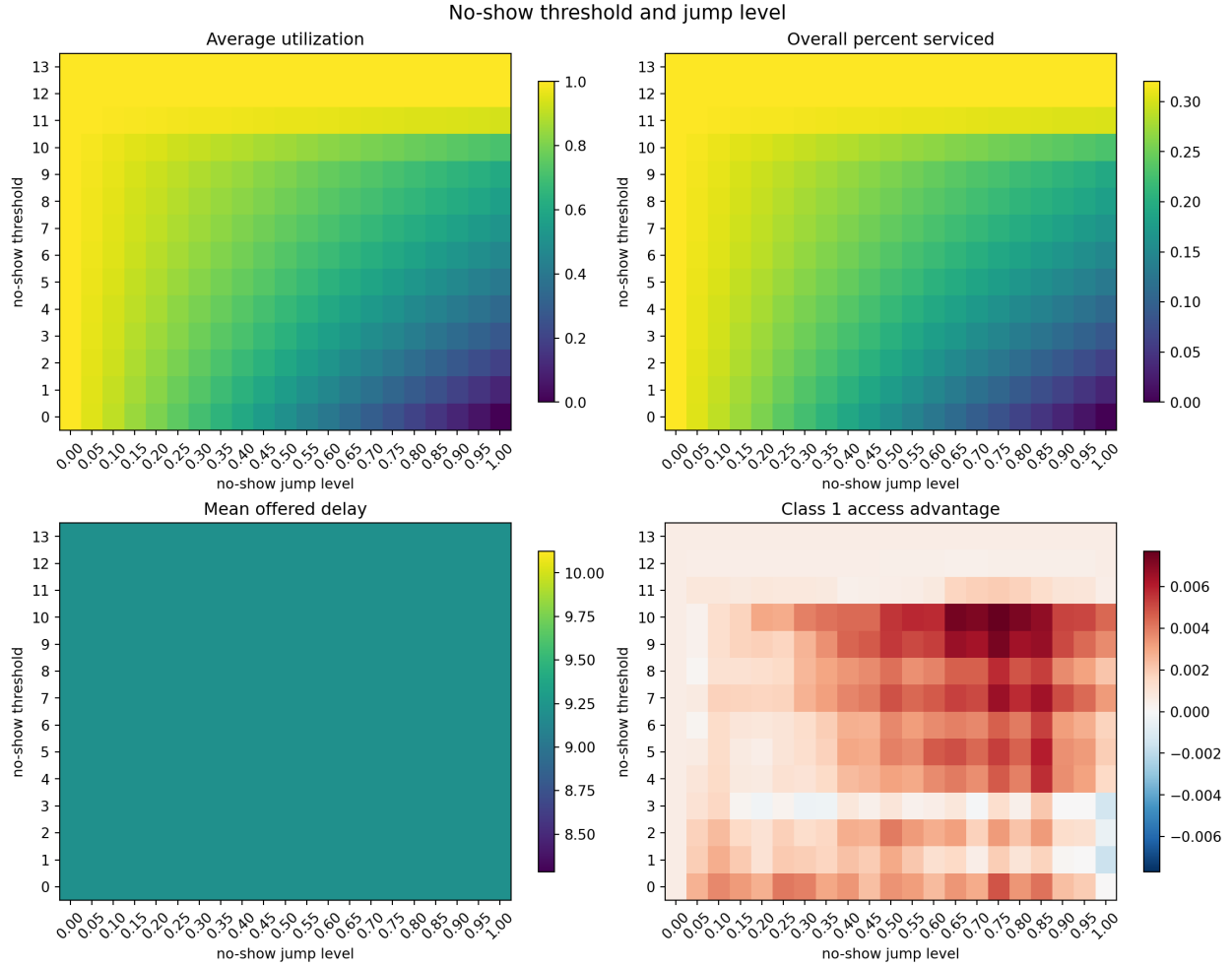


Figure 10: No-show sensitivity when both classes share the same threshold and jump level.

This sweep varies the absolute no-show threshold and the absolute no-show jump level together. Both patient classes use the same values at each grid point.

Aggregate impact. This is a strong utilization sweep. Low thresholds combined with high jump levels cause many booked appointments to become no-shows, reducing average utilization and overall percent serviced. High thresholds or low jump levels preserve completed visits.

Class-benefit dynamic. Since both classes share the same no-show rule, there is almost no structural class advantage. The largest observed access advantage is about 0.008 in absolute value. The delay panels also remain mostly unchanged because no-show behavior happens after booking.

Interpretation. Absolute no-show threshold and jump level explain utilization loss. Class benefit appears only when one class has a lower no-show jump or a higher no-show threshold than the other.

7 Cancellation

7.1 Probability Sensitivity

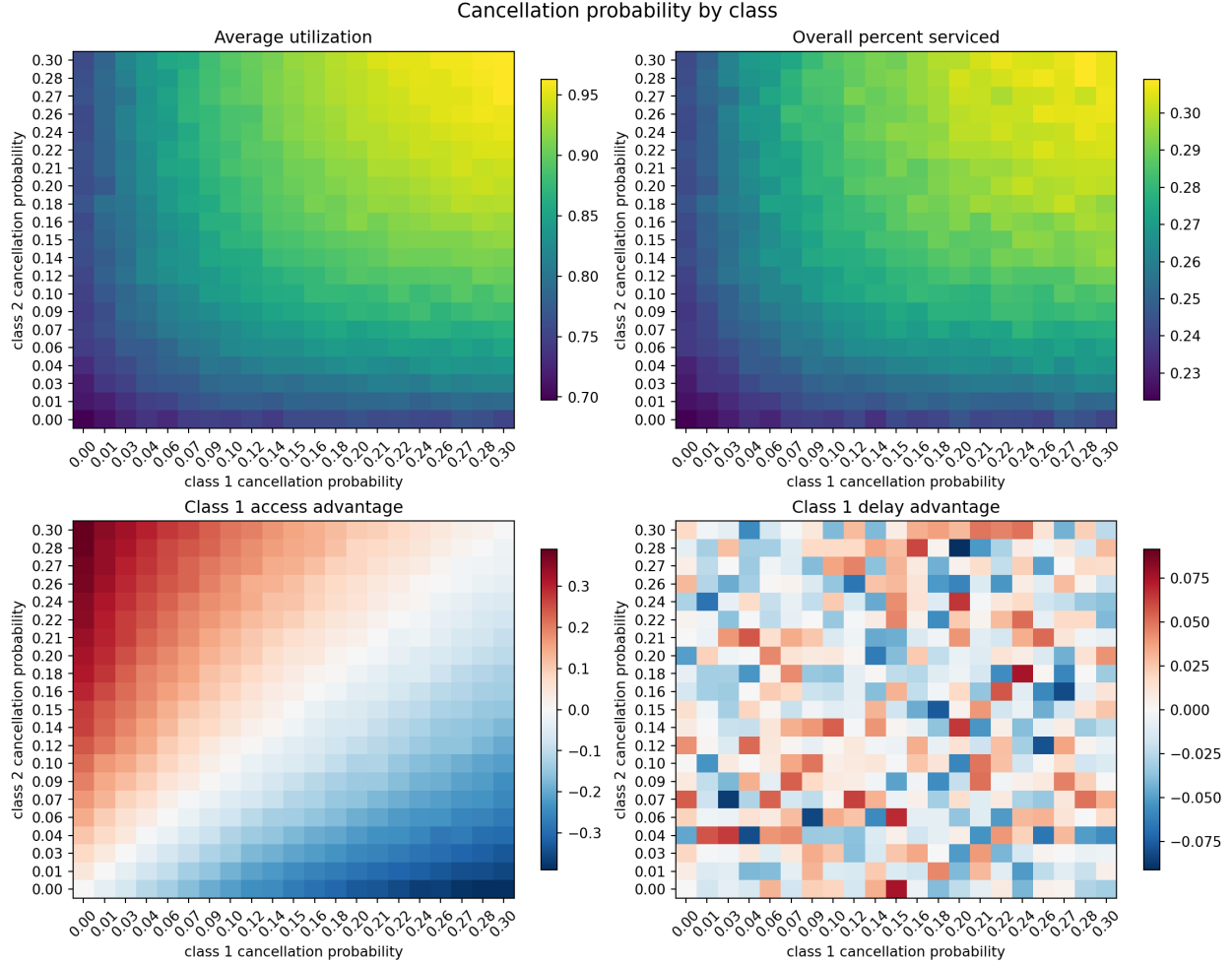


Figure 11: Cancellation probability sensitivity by class.

Cancellation is not threshold-based in the current model. It is a constant daily probability for future appointments.

Aggregate impact. Cancellation has a large effect on access and also changes capacity availability dynamically. Canceled appointments free future capacity, but the canceling patient is lost unless they arrive again later as a new patient.

Class-benefit dynamic. Cancellation asymmetry is the strongest class-benefit lever in this analysis. The class with the lower cancellation probability is much more likely to complete service. The largest access advantage is about 0.39 in absolute value, larger than the class advantage from balking or no-show parameters.

Interpretation. If cancellation differs by class, class outcomes can diverge sharply even when arrival rates and booking rules are otherwise symmetric.

8 Arrival and Demand

8.1 Arrival Rate and Class Mix

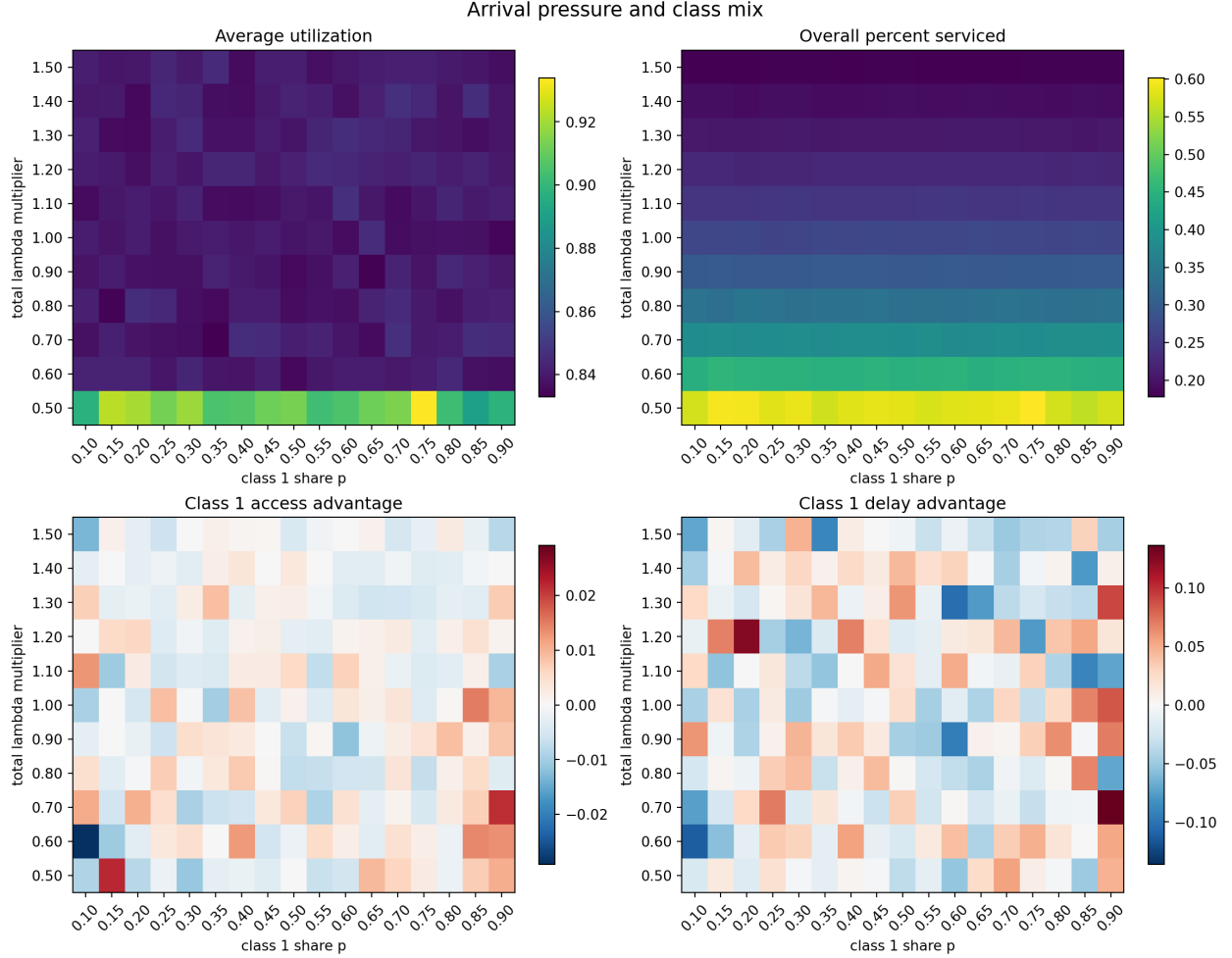


Figure 12: Total arrival-rate and class-mix sensitivity.

This sweep varies:

$$\lambda_{\text{total}} = \text{multiplier} \times \lambda_{\text{total, baseline}},$$

$$\lambda_1 = p\lambda_{\text{total}}, \quad \lambda_2 = (1 - p)\lambda_{\text{total}}.$$

Aggregate impact. Total demand pressure matters strongly. As total arrival rate increases, overall percent serviced decreases and offered delay increases. Average utilization can remain high even when access is poor.

Class-benefit dynamic. Class mix has much weaker class-benefit effects than behavioral asymmetry. In the fine grid, the largest access advantage from arrival mix is about 0.029 in absolute value. This is small compared with cancellation, balking, or no-show asymmetries.

Interpretation. Because the two classes have identical behavior and value in the baseline, changing p mostly changes volume, not priority. Class mix alone does not create a strong structural benefit.

8.2 Class Arrival Rate Sweeps

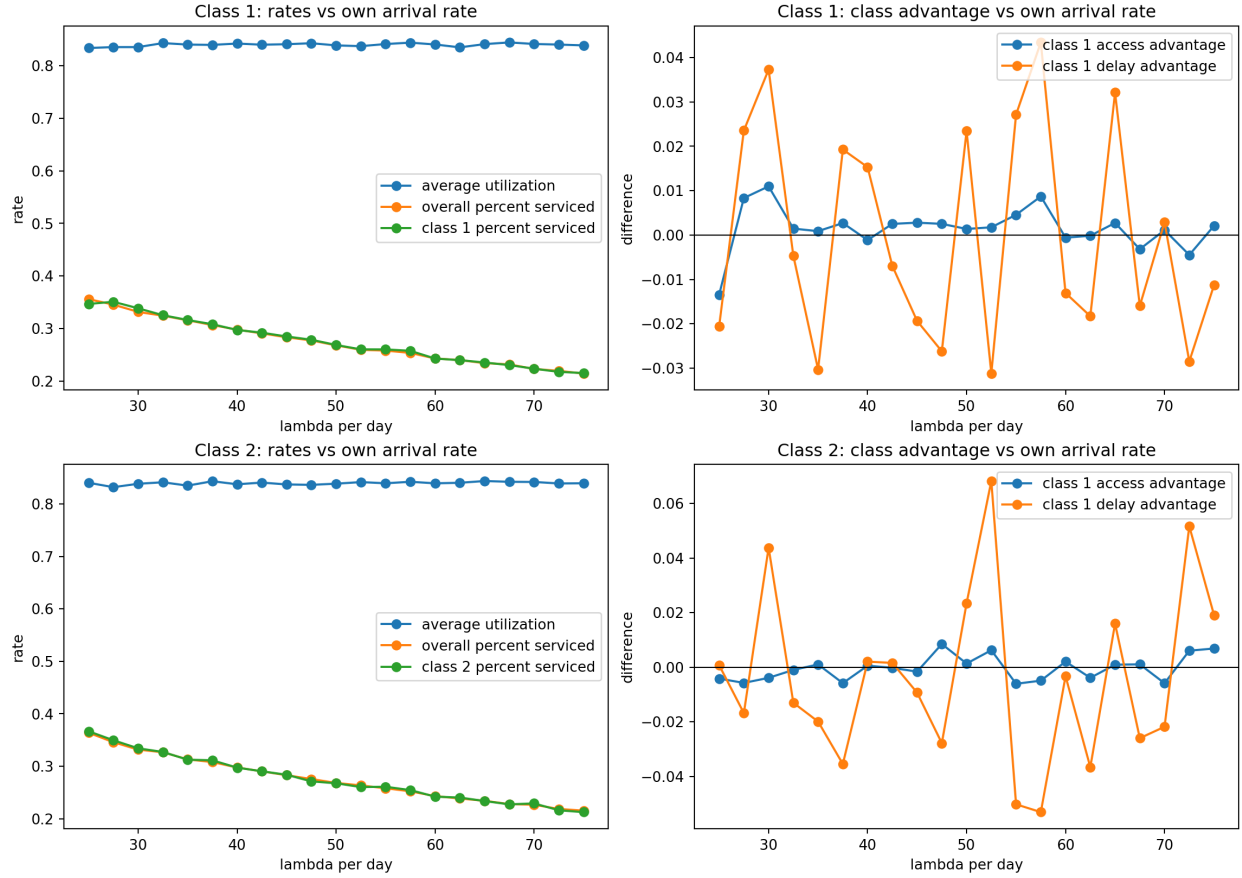


Figure 13: Class-specific arrival-rate sweeps.

These plots vary one class's arrival rate at a time while holding the other class fixed.

Aggregate impact. Increasing either class's arrival rate increases congestion. Percent serviced decreases and booking delays increase once the system is capacity constrained.

Class-benefit dynamic. Changing only one class's arrival rate does not create a large persistent class advantage in the current symmetric baseline. The largest access advantage in this sweep is about 0.014 in absolute value, and the largest delay advantage is about 0.068 days.

Interpretation. Own-class arrival volume matters for congestion, but because the booking policy is FCFS and the classes are behaviorally symmetric, it does not strongly privilege either class.

8.3 Additional FCFS Policy Analysis

The previous sections vary many model parameters. This section focuses only on the first policy, FCFS, and asks how it behaves as total demand increases while the two classes remain symmetric.

8.3.1 Code Used

The figure generator now includes a dedicated FCFS stress loop. The full code is in `docs/generate_metric_analysis_figures.py`; the core logic is:

```
lambda_total_base = sum(
    p.lambda_per_day for p in BASE_CONFIG.classes.values()
)
class_1_share = (
    BASE_CONFIG.classes[1].lambda_per_day / lambda_total_base
)

for multiplier in FCFS_STRESS_MULTIPLIERS:
    lambda_total = lambda_total_base * multiplier
    config = set_arrival_mix(BASE_CONFIG, lambda_total, class_1_share)
    result = run_result(config, seed)
    rows.append({
        "arrival_multiplier": multiplier,
        **result_metrics_from_result(result),
        **outcome_rates_from_result(result),
    })
```

The outcome decomposition is measured relative to arrivals:

$$1 = r_{\text{served}} + r_{\text{balked}} + r_{\text{no offer}} + r_{\text{canceled}} + r_{\text{no-show}} + r_{\text{unresolved booked}}.$$

8.3.2 Capacity Stress Results

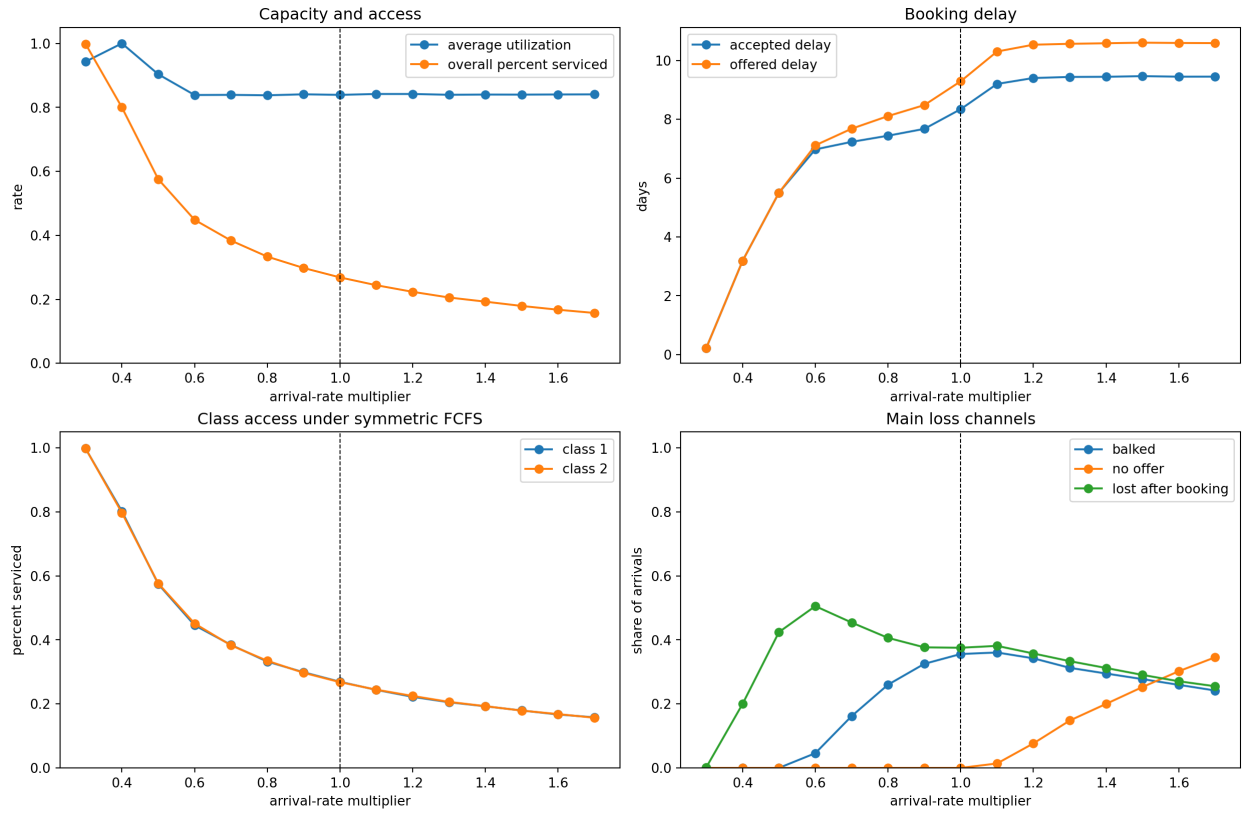


Figure 14: FCFS capacity stress curves as total arrival rate changes and class mix stays 50/50.

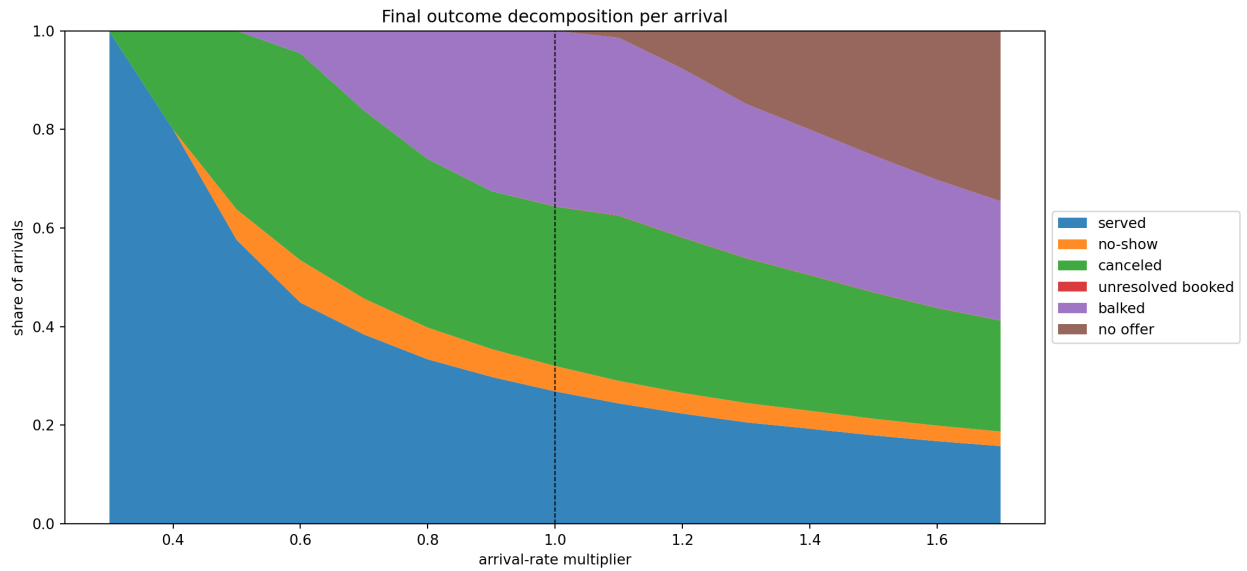


Figure 15: Final outcome decomposition for measured arrivals under FCFS.

Arrival mult.	Avg. util.	Overall svc.	Offered delay	Served	Balked	No offer	Lost after booking	Δ_{access}
0.30	0.943	0.998	0.22	0.998	0.000	0.000	0.002	-0.000
1.00	0.839	0.269	9.30	0.269	0.356	0.000	0.376	0.001
1.70	0.841	0.157	10.59	0.157	0.242	0.346	0.255	0.001

Table 5: Selected FCFS stress points. Outcome columns are shares of measured arrivals. Lost after booking combines cancellations, no-shows, and unresolved booked patients at the end of the run.

Aggregate impact. FCFS has a clear stress pattern. When demand is below capacity, almost every arrival is served, and utilization is below 1 only because there are not enough arrivals to fill every slot. Near the baseline load, utilization remains high, but only about 27% of arrivals complete service. Under heavy overload, utilization is still about 84%, while the served share falls to about 16%.

Failure-mode dynamic. The loss channel changes with load. At baseline, there is almost no no-offer loss; patients mostly leave through balking or after-booking losses. At high arrival pressure, no-offers become a major failure mode because the FCFS calendar horizon fills.

Class-benefit dynamic. This FCFS stress test preserves identical behavior for both classes. The largest observed access gap across the stress curve is about 0.005, so congestion alone does not create a meaningful class benefit. It hurts both classes almost equally.

Interpretation. Utilization is not enough to evaluate FCFS. The clinic can look busy while access is poor. The best aggregate diagnostic pair is `average_utilization` with `overall_percent_served`, plus the outcome decomposition to show whether patients are lost before booking, at offer time, or after booking.

9 Regression Screening Analysis

The heatmaps show one or two parameters at a time. To compare many parameters simultaneously, I also ran a randomized FCFS simulation screen and fit standardized linear regressions to the simulated outputs.

9.1 Design

- 240 random parameter settings were sampled.
- Each setting was simulated with 2 random seeds and averaged.
- The regression inputs were total arrival rate, class mix, and mean/gap versions of balking, no-show, and cancellation parameters.
- Coefficients are standardized: larger absolute values mean stronger average association over the sampled range.

For example:

$$\text{balking step mean} = \frac{\text{class 1 step} + \text{class 2 step}}{2}, \quad \text{balking step gap} = \text{class 1 step} - \text{class 2 step}.$$

Mean features mainly explain aggregate outcomes. Gap features mainly explain which class is benefited.

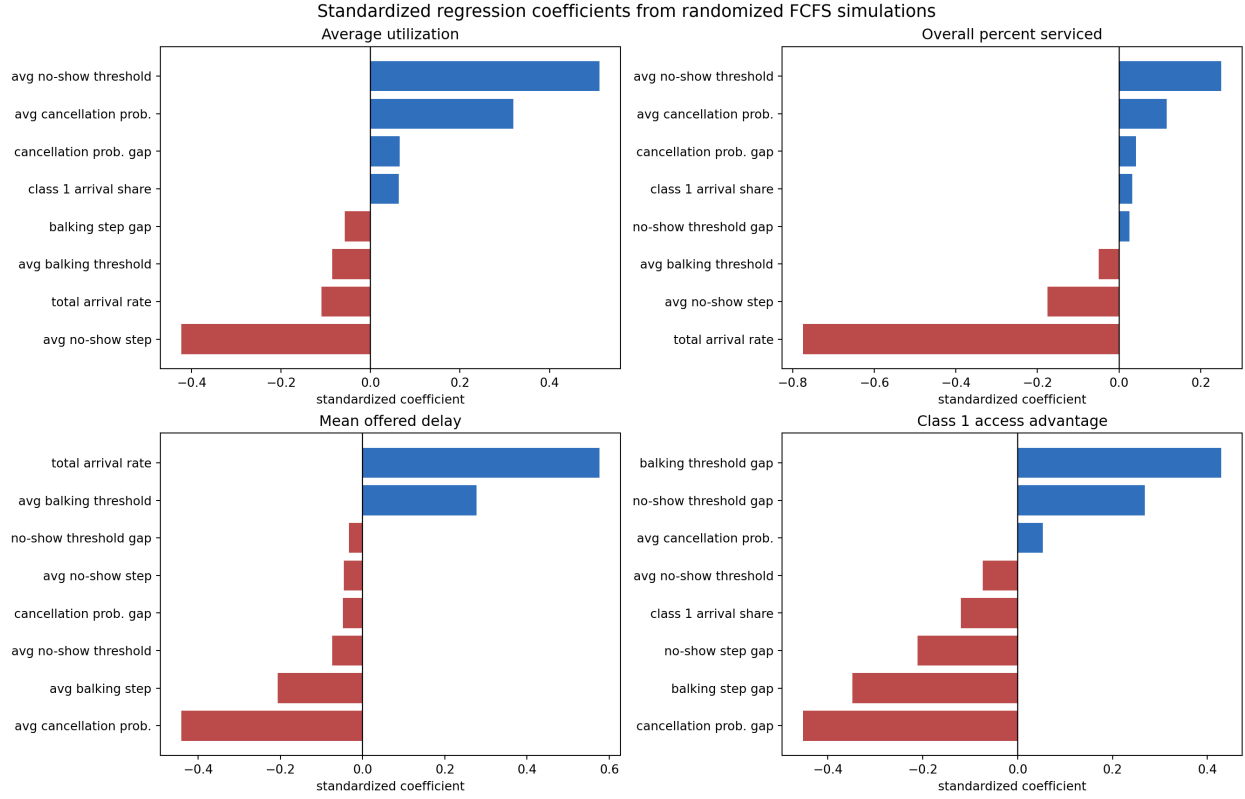


Figure 16: Largest standardized regression coefficients from the randomized FCFS simulation screen. Positive coefficients increase the target metric; negative coefficients decrease it.

Target metric	Full-sample R^2	Test R^2
Average utilization	0.675	0.545
Overall percent serviced	0.799	0.719
Mean offered delay	0.718	0.555
Class 1 access advantage	0.639	0.708

Table 6: Regression fit quality. Test R^2 uses a random 80/20 train/test split.

Target	Most important feature	Standardized coefficient
Average utilization	Average no-show threshold	0.512
Average utilization	Average no-show step	-0.423
Average utilization	Average cancellation probability	0.320
Overall percent serviced	Total arrival rate	-0.774
Overall percent serviced	Average no-show threshold	0.251
Overall percent serviced	Average no-show step	-0.175
Mean offered delay	Total arrival rate	0.576
Mean offered delay	Average cancellation probability	-0.441
Mean offered delay	Average balking threshold	0.278
Class 1 access advantage	Cancellation probability gap	-0.452
Class 1 access advantage	Balking threshold gap	0.429
Class 1 access advantage	Balking step gap	-0.349

Table 7: Top regression drivers. Gap means class 1 value minus class 2 value.

Aggregate findings. Overall access is dominated by demand pressure: higher total arrival rate has the largest negative coefficient for `overall_percent_served`. Utilization is driven mostly by no-show behavior, especially the no-show threshold and step. Offered delay increases with total arrival rate and with more tolerant balking thresholds.

Class-benefit findings. Class advantage is explained by class differences, not by absolute parameter levels. Class 1 is hurt when its cancellation probability is higher than class 2's, helped when its balking threshold is higher, hurt when its balking step is higher, helped when its no-show threshold is higher, and hurt when its no-show step is higher.

Caution. This is a screening regression, not a replacement for the heatmaps. The model is nonlinear, so coefficients should be read as average directional effects over the sampled range. The CSV outputs are saved as `regression_simulation_data.csv`, `regression_standardized_coefficients.csv`, and `regression_model_scores.csv`.

10 Impact Ranking

Parameter sweep	Max $ \Delta_{\text{access}} $	Main aggregate effect	Main class-benefit interpretation
Cancellation probability	0.390	Access and capacity dynamics	Lower cancellation class is strongly benefited.
Balking threshold	0.178	Delay/access tradeoff	Higher tolerance class is benefited.
Balking step	0.174	Delay/access tradeoff	Lower balking class is benefited.
No-show step	0.172	Utilization	Lower no-show class is benefited in completion, not delay.
No-show threshold	0.098	Utilization	Higher threshold class is benefited.
Arrival mix	0.029	Demand pressure	Weak class-benefit effect under symmetric behavior.
Own-class arrival rate	0.014	Congestion	Weak class-benefit effect under FCFS and symmetric behavior.
Common balking threshold-jump	0.014	Delay/access tradeoff	Absolute effect; no structural class benefit.
Common no-show threshold-jump	0.008	Utilization	Absolute effect; no structural class benefit.

Table 8: Relative strength of class-benefit effects in the fine-grid analysis.

11 Summary of What Matters

1. **For aggregate utilization, no-show behavior matters most.** No-shows directly waste booked slots.
2. **For aggregate access and delay, total demand pressure matters strongly.** Higher arrival pressure lowers percent serviced and increases offered delay.
3. **For FCFS specifically, high utilization can hide poor access.** In the stress curve, utilization stays near 0.84 at high load even while the served share falls to about 0.16.
4. **The regression screen confirms the same split.** Aggregate performance is driven mainly by total arrival rate and average no-show behavior; class advantage is driven by class-specific gaps in cancellation, balking, and no-show parameters.
5. **For class benefit, asymmetry matters more than absolute level.** Off-diagonal heatmap regions show which class is benefited. The diagonal and the common threshold-jump sweeps show the absolute-level effect when both classes are treated similarly.
6. **Cancellation asymmetry is the largest class-benefit driver.** A class with lower cancellation probability has much higher percent serviced.
7. **Balking asymmetry controls who stays in the system.** The class that balks less, or tolerates longer waits, is served more often.
8. **No-show asymmetry controls who converts bookings into completed visits.** It changes percent serviced and utilization, but barely changes offered booking delay.
9. **Arrival mix alone is not enough to create strong class advantage in this baseline.** Since classes are behaviorally identical, class mix mostly changes demand pressure.